

CS336 Assignment 1 Writeup

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2026-02-12

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1 Unicode、BPE 与 Tokenizer 题目

1.1 Problem unicode1 与 unicode2

1. `chr(0)` 返回 ASCII 的 NUL (`U+0000`)。
2. `repr(chr(0))` 显示其转义形式 (如 `"\x00"`)，而 `print(chr(0))` 不可见。
3. NUL 在字符串中通常作为不可见控制字符存在，很多显示与处理链路会忽略或特殊处理。
4. UTF-8 相比 UTF-16/UTF-32 在网页文本上更紧凑、生态兼容最好，适合字节级 tokenizer。
5. 错误 UTF-8 解码函数逐字节解码会破坏多字节字符，例如 `b'\xe7\x89\xb'` (“牛”) 会失败或错误。
6. 非法两字节序列示例: `b'\xc0\xaf'`

1.2 BPE 训练结果与分析 (Problem train_bpe_*)

1.2.1 TinyStories 10k

训练耗时: 36.58s, 峰值内存: 254.65 MB, 最长 token: ' accomplishment' (15 bytes, 含前导空格)

该最长 token 语义自然, BPE 在该语域下学到的是高频可解释子词。

1.2.2 OWT 32k

训练耗时: 836.67s, 峰值内存: 17 777.30 MB, 最长 token 长度 64 bytes, 内容为:
AA

这是网页语料的噪声 (可能是字符串解码错误) 造成的。

1.2.3 训练瓶颈说明

通过 Profile, BPE 训练主耗时来自预分词与 pair 计数维护。分别对应 `_pretokenize_parallel` 和 `pair_counts/pair_occurrences` 与 `_update_pair`。

1.3 Tokenizer 实验 (Problem tokenizer_experiments)

- 将 TinyStories tokenizer 用于 TinyStories 的压缩率: 4.248 bytes/token
- 将 OWT tokenizer 用于 OWT 的压缩率: 4.682 bytes/token
- OWT 样本用 TinyStories tokenizer: 3.261 bytes/token, 我们发现压缩率相比用自身 tokenizer 降低了。
- OWT valid 编码吞吐: 7.25 MiB/s
- TinyStories valid 编码吞吐: 10.90 MiB/s

Pile 耗时估算：

$$t \approx \frac{825 \times 1024^3}{7.6011956 \times 10^6} \approx 1.165 \times 10^5 \text{ s} \approx 1.35 \text{ days}.$$

为什么用 uint16: 词表规模最大 32k, token id 全在 $[0, 65535]$ 内, uint16 足够且高效。

2 计算题详细推导

2.1 Problem transformer_accounting

2.1.1 参数量推导 (GPT-2 XL-shaped)

记: 词表大小 V 、层数 L 、隐藏维度 d 、FFN 维度 d_{ff} 。本实现采用 SwiGLU (三矩阵), 且 embedding 与 lm_head 不绑权。

参数分解

$$P_{emb} = Vd, \quad (1)$$

$$P_{head} = Vd, \quad (2)$$

$$P_{attn/layer} = 4d^2 \quad (Q, K, V, O), \quad (3)$$

$$P_{ffn/layer} = 3dd_{ff} \quad (W_1, W_2, W_3), \quad (4)$$

$$P_{norm/layer} = 2d, \quad (5)$$

$$P_{norm,final} = d. \quad (6)$$

总参数:

$$P = 2Vd + L(4d^2 + 3dd_{ff} + 2d) + d.$$

代入 $V = 50257, L = 48, d = 1600, d_{ff} = 6400$:

$$P = 2,127,057,600.$$

float32 参数内存:

$$M_{param} = 4P \approx 7.92 \text{ GiB}.$$

2.1.2 前向 FLOPs 推导

设 batch 为 B , 上下文长度为 T 。使用矩阵乘规则: $A \in \mathbb{R}^{m \times n}, B \in \mathbb{R}^{n \times p} \Rightarrow 2mnp$ FLOPs。

单层 FLOPs

$$F_{QKV} = 3 \cdot 2BTd^2 = 6BTd^2, \quad (7)$$

$$F_{attn\ mat} = 2BT^2d(QK^\top) + 2BT^2d(AV) = 4BT^2d, \quad (8)$$

$$F_{out\ proj} = 2BTd^2, \quad (9)$$

$$F_{FFN} = 2BTdd_{ff} + 2BTdd_{ff} + 2BTd_{ff}d = 6BTdd_{ff}. \quad (10)$$

最终输出头:

$$F_{head} = 2BTdV.$$

总 FLOPs:

$$F = L(8BTd^2 + 4BT^2d + 6BTdd_{ff}) + 2BTdV.$$

代入 GPT-2 XL-shaped, $B = 1, T = 1024$:

$$F \approx 4.513 \times 10^{12}.$$

2.1.3 FLOPs 占比与规模变化

表 1: 固定 $T = 1024, B = 1$ 时, 不同模型 FLOPs 组成占比

模型	Attention 投影	Attention 矩阵乘	FFN	LM Head
gpt2-small	16.58%	11.06%	49.75%	22.61%
gpt2-medium	19.96%	9.98%	59.87%	10.20%
gpt2-large	21.40%	8.56%	64.20%	5.84%
gpt2-xl	22.30%	7.14%	66.91%	3.65%

结论: 模型越大 (固定 T), FFN 与投影占比上升, T^2 注意力和 LM Head 占比下降。

2.1.4 上下文扩展到 T=16384

GPT-2 XL-shaped 时:

$$F_{16384} \approx 1.495 \times 10^{14}, \quad \frac{F_{16384}}{F_{1024}} \approx 33.13.$$

由于 T^2 项增长, Attention 矩阵乘占比升至约 55.15%, 成为主导项。

2.2 AdamW Accounting (按题面简化口径)

题面口径使用 $d_{ff} = 4d$ 且 FFN 记作 $W_1\text{-SiLU-}W_2$ 。

2.2.1 内存分解

参数项

$$P = 2Vd + L(12d^2 + 2d) + d.$$

$$M_{param} = 4P, \quad M_{grad} = 4P, \quad M_{opt} = 8P.$$

激活项 (上界近似) 按题面列出的组件进行近似汇总:

$$A \approx L(16BTd + 2BhT^2) + BTd + 2BTV,$$

$$M_{act} \approx 4A.$$

故峰值近似:

$$M_{total} \approx M_{param} + M_{grad} + M_{opt} + M_{act} = 16P + 4A.$$

2.2.2 代入 GPT-2 XL-shaped

取 $V = 50257, T = 1024, L = 48, d = 1600, h = 25$, 得到

$$M_{total}(B) \approx 15,517,753,344 \cdot B + 26,168,601,600 \quad (\text{bytes}),$$

即

$$M_{total}(B) \approx 14.452B + 24.371 \quad (\text{GiB}).$$

$$B_{max} = \lfloor \frac{80-24.371}{14.452} \rfloor = 3, \text{ 所以 } 80\text{GiB 下最大 batch 约 } 3.$$

2.2.3 AdamW 单步 FLOPs 与训练时长

单步 FLOPs 近似:

$$F_{step} \approx 3F_{forward} + cP$$

其中 c 为常数 (参数更新项), 主导仍是前向 + 反向。

按 A100 float32 峰值 19.5 TFLOP/s、MFU=50%、400k steps、batch=1024:

$$\text{days} \approx 5115.$$

数量级结论: 单卡 float32 下该规模训练不现实。

3 训练实验与图表

3.1 学习率扫描与发散证据 (TinyStories)

表 2: Learning Rate Sweep (TinyStories)

lr	报告 val loss	备注
1e-4	2.1700	稳定收敛
3e-4	1.7620	稳定收敛
1e-3	1.5593	最优区间
3e-3	1.6799	开始退化
1e-2	2.5253	明显退化
3e-2	3.7422	高损失不稳定
1e-1	4.4049	高损失不稳定
3e-1	4.2369	出现尖峰 (max val=33.63)
1e0	57.5486 (末次)	发散 (max val=149.86)
3e0	117.5958 (末次)	发散 (max val=876.86)

主实验 (ts_main_lr1e3_20k) 达到

$$\text{best val loss} = 1.3730 (< 1.45).$$

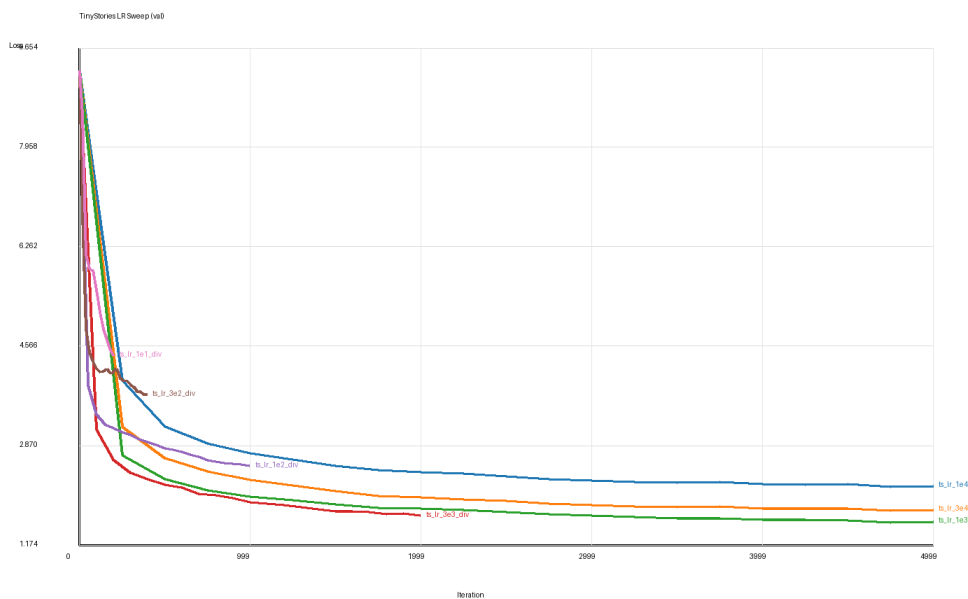


图 1: TinyStories 学习率扫描 (验证集)

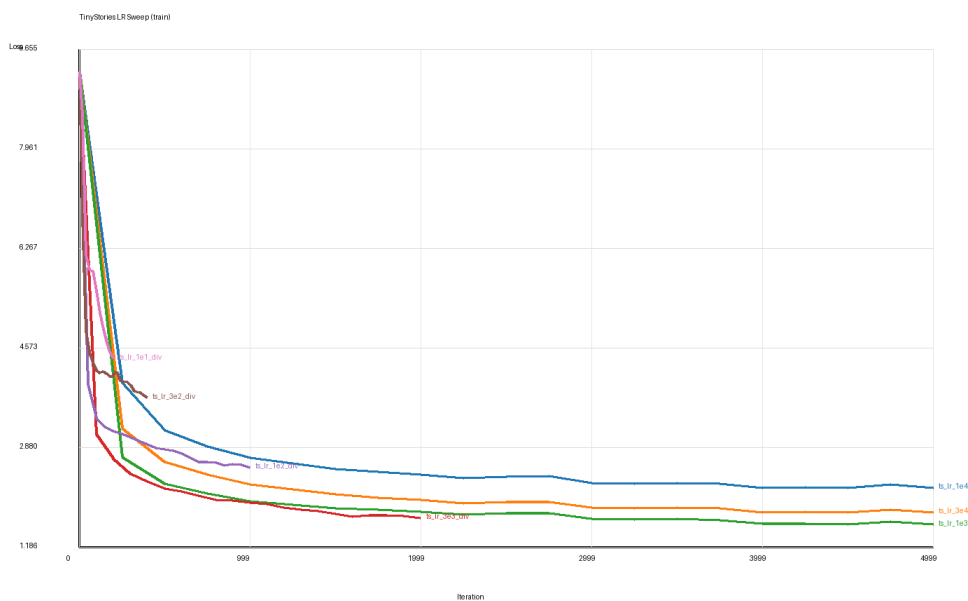


图 2: TinyStories 学习率扫描 (训练集)

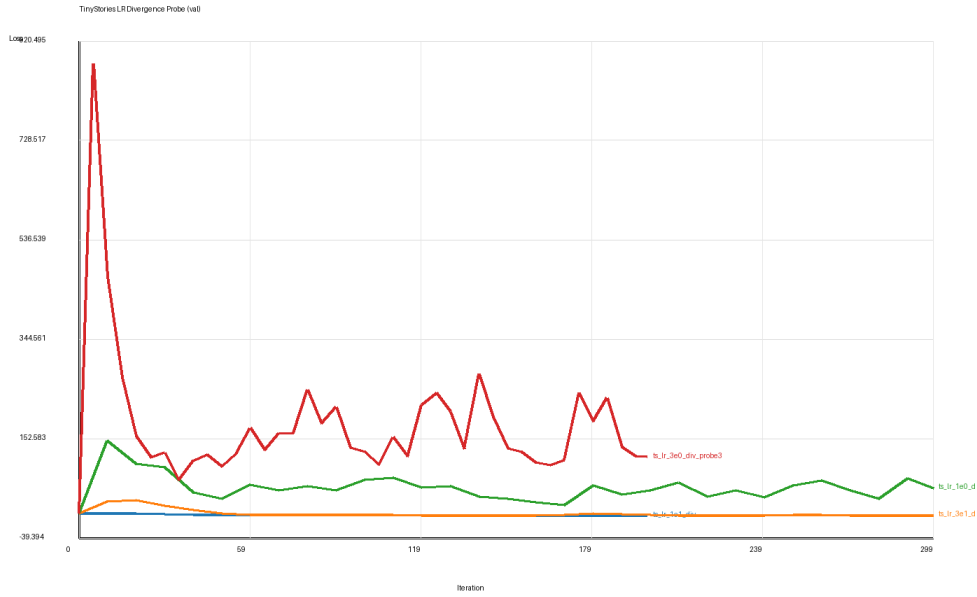


图 3: 高学习率发散探测 (验证集)

发散判据采用验证损失相对初始值的显著放大。补跑结果如下:

- $lr=1.0$: 末次 val loss=57.5486; 峰值 $149.86/9.27 \approx 16.17$ 倍初始值
- $lr=3.0$: 末次 val loss=117.5958; 峰值 $876.86/9.27 \approx 94.61$ 倍初始值

对应日志: `slurm_logs/train_lm_1478.out`、`slurm_logs/train_lm_1479.out`。

3.2 Batch Size 实验

表 3: batch size 从 1 到显存上限探测

batch	lr	status	best val loss	tokens seen	备注
1	3e-4	ok	2.9371	1.28M	可训练
32	1e-3	ok	1.6665	40.96M	可训练
64	1e-3	ok	1.5593	81.92M	可训练
128	1.5e-3	ok	1.4614	163.84M	可训练
256	1.5e-3	ok	2.2109	19.66M	显存仍可容纳 (300 iter 探测)
512	1.5e-3	oom	-	-	首个 forward 即 OOM

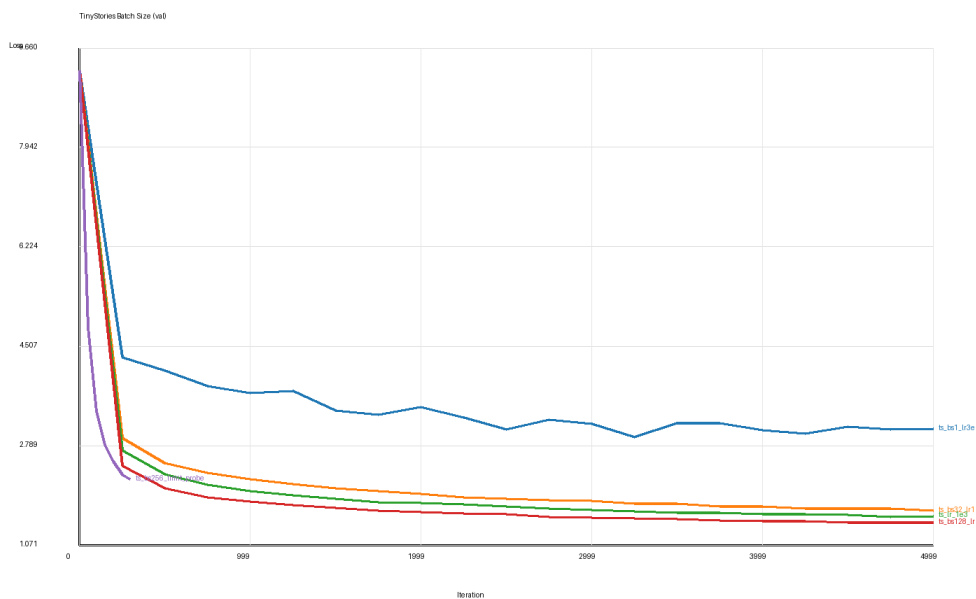


图 4: Batch size 对比 (同训练步数)

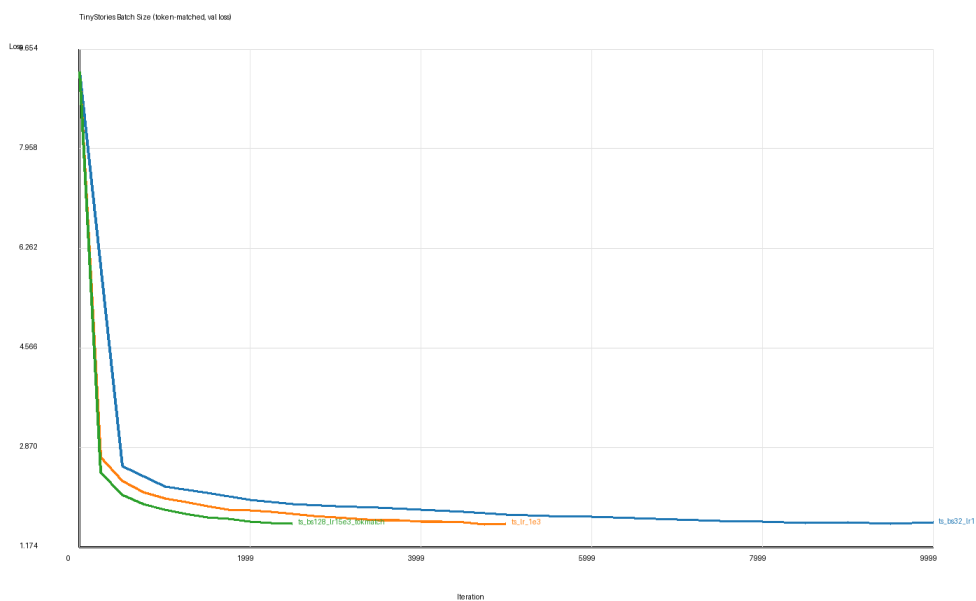


图 5: Batch size 对比 (token-matched)

因此在当前 32GB 显存与该模型配置下，最大可运行 batch 位于 $256 < B_{\max} < 512$ 。OOM 证据见 `slurm_logs/train_lm_1475.err`。

3.3 消融实验

表 4: TinyStories 消融 (5k steps)

设置	best val loss
baseline (pre-norm + RMSNorm + RoPE + SwiGLU)	1.5593
去 RMSNorm (lr=1e-3)	1.5773
去 RMSNorm (lr=3e-4)	1.7679
post-norm	1.5553
NoPE (去 RoPE)	1.6537
SiLU (替换 SwiGLU, d_ff=2048)	1.5835

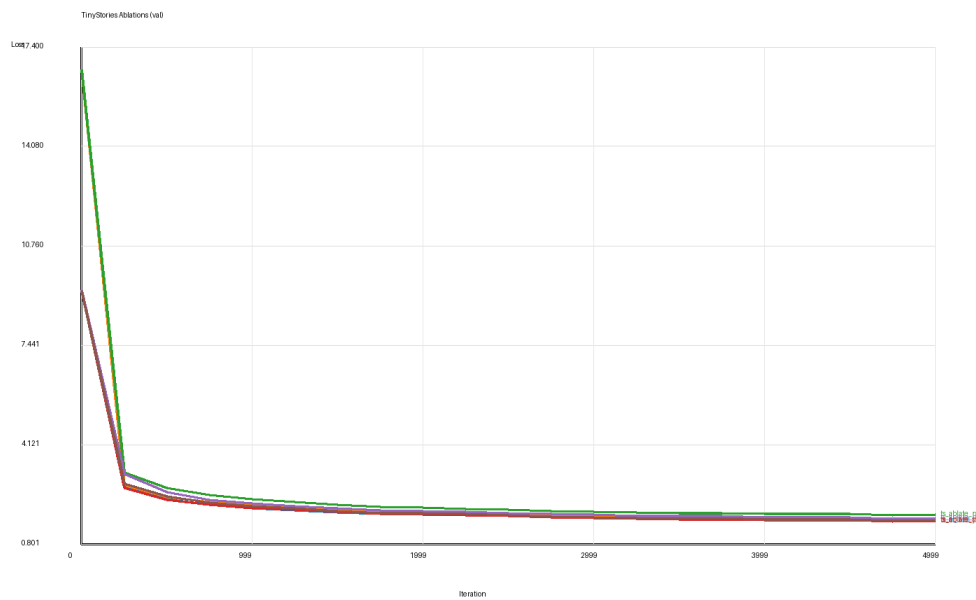


图 6: TinyStories 各消融验证曲线

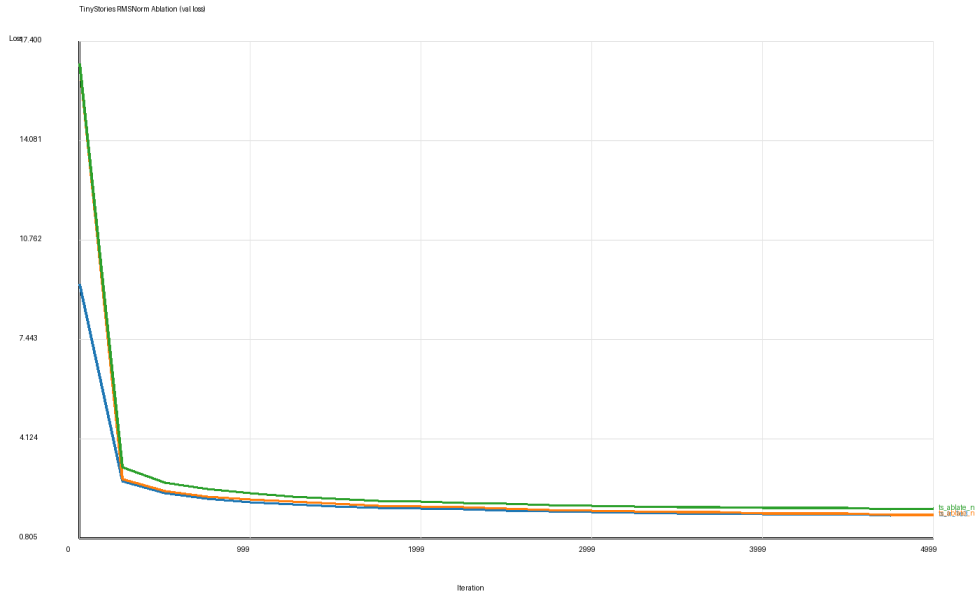


图 7: RMSNorm 消融细节曲线

本次 SiLU 消融按 handout 要求做参数量匹配： $d_{ff} = 4 \times d_{model} = 4 \times 512 = 2048$ 。

3.4 OpenWebText 主实验

- TinyStories 主实验 best val loss: 1.3730
- OWT 主实验 best val loss: 4.0370

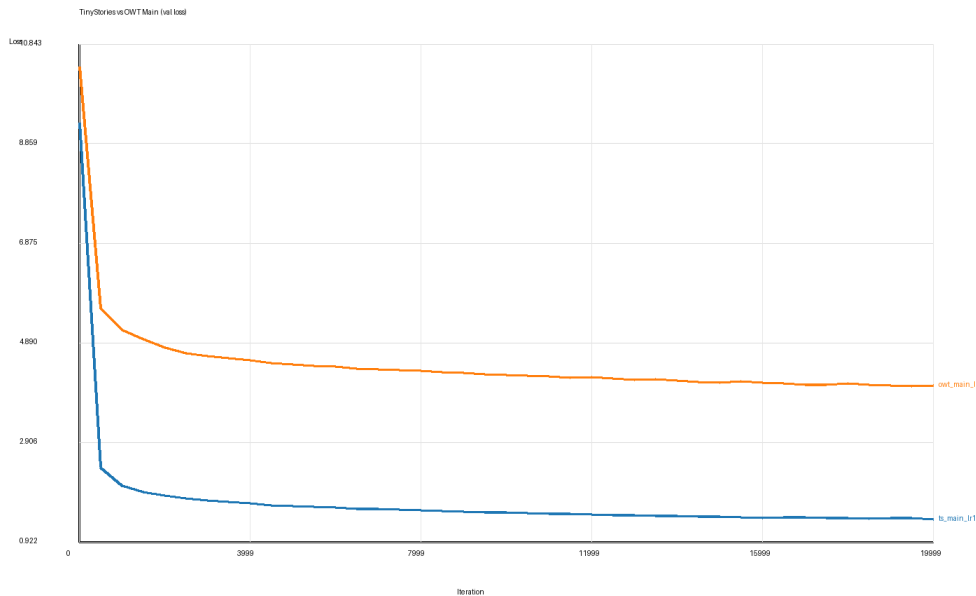


图 8: TinyStories 与 OWT 主实验验证曲线对比

解释：OWT 数据分布更复杂、熵更高，在同模型规模与 token 预算下拟合不足，因此 loss 更高，生成质量更弱。

4 生成样例与质量分析

4.1 TinyStories 生成（满足 EOT 终止）

```
1 Once upon a time, there was a little dog named Spot. Spot was a happy dog who
  loved to play with his friends. One day, Spot found a big ball in the yard.
  He was so happy and wanted to play with it.
2 Spot decided to share the ball with his friends. He called his friends over
  and said, "Let's play!" But when his friends came, they all wanted to play
  with the ball too. This made Spot and his friends very sad. They did not
  want to share their ball.
3 Spot had an idea. He said, "Let's split the ball with our hands!" His friends
  thought it was a great idea. They all worked together and split the ball.
  Spot and his friends were so happy. They played with the big ball all day
  long and had lots of fun.<|endoftext|>
```

Listing 1: TinyStories 生成样例（满足 EOT 终止）

该样例编码长度 169 token，并在首个 <|endoftext|> 终止，满足作业要求。

4.2 OWT 生成（满足至少 256 token）

```
1 The history of machine learning, and even that has been a significant part of
  our time. The world has been a lot of fun. The game has been played on over
  the past several years, and it's been a wonderful experience.
2
3 My first experience with the game was, with the people playing on social media
  , and I was there, and I was very happy about it. I enjoyed this game. All
  that brings to the attention of the game. From my perspective you're not
  going to be a champion. You're going to be in that game, and I hope you
  feel something.
4
5 What about the community?
6
7 I have always been fascinated with the game. We're always looking forward to
  that. I've always felt that it is a challenge to the industry and that my
  fans like to see it are being used for anything that comes from the game.
  We are just trying to make games that you never think. You just want the
  game to be a part of it.
8
9 You love that they did some really good things. If you want to develop a game,
  you're really good. A lot of the games you have, it's a really good
  experience.
```

Listing 2: OWT 生成样例

该样例编码长度 262 token，满足“至少 256 token”要求。

4.3 影响生成质量的因素

1. 数据复杂度：OWT 的主题与词汇更复杂，难度显著高于 TinyStories。
2. 模型容量与训练预算：17M 级模型在固定 token 预算下对 OWT 表达能力不足。
3. 解码超参：temperature/top-k/top-p 会显著改变流畅度与多样性。